



RwaCow — Rwanda Cow Tracking System

A full-stack web application for registering, tracking, and managing cattle across Rwanda. It covers the complete lifecycle of a cow: registration, ownership, location, vaccination, offspring (*izakomotse*), sale/transfer, and slaughter (*izabazwe*).

Available in **English** and **Kinyarwanda**, with role-based access for farmers, veterinarians, government officials, slaughterhouses, and administrators.



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What the System Does

Feature	Description
 Cow Registration	Record each cow with a unique tag number, breed, gender, birth date, color, and weight
 Owner Management	Track owners by national ID, phone, and location
 Location Tracking	Locate each cow by Province → District → Sector → Cell → Village (+ optional GPS)
 Vaccination Records	Record vaccines given, the date, and the next due date
 Offspring (Izakomotse)	Register calves born from a cow, linked to their mother
 Slaughter (Izabazwe)	Schedule and confirm slaughter, record meat weight
 Ownership Transfer	Re-register a cow to a new owner when sold, with full history
 Bilingual	Full English + Kinyarwanda interface
 Role-Based Access	Each role sees and does only what it should
 Excel Export & Backup	Download data as Excel; admins can back up the whole database

Who Uses It (Roles)

Role	Kinyarwanda	What they can do
Admin	Umuyobozi Mukuru	Full access to everything + full database backup
Government	Umukozi wa Leta	View all data, manage owners, export reports
Veterinarian	Muganga w'Amatungo	Record vaccinations, update cow health info
Farmer	Umworozi	Manage only their own cows, register offspring, transfer ownership
Slaughterhouse	Ubwicanyi bw'Amatungo	Schedule and confirm slaughter of cattle

Technology Stack

Layer	Technology
Frontend	React 18 + TypeScript + Vite + Tailwind CSS
Backend	Node.js + Express
Database	MySQL (MariaDB via XAMPP) — with SQLite fallback
Authentication	JWT (JSON Web Tokens) + bcrypt password hashing
Excel Export	SheetJS (xlsx)

System Requirements

- **Node.js** 18 or newer ([download](#))
- **XAMPP** (for MySQL/MariaDB) ([download](#))
- A modern web browser (Chrome, Edge, Firefox)
- Windows, macOS, or Linux

Installation & Setup

Step 1 — Start MySQL

- Open the **XAMPP Control Panel**
- Click **Start** next to **MySQL**
- Wait until it turns green (running on port 3306)

Step 2 — Configure the backend

The backend reads settings from `backend/.env`. The defaults match XAMPP:

```
DB_ENGINE=mysql
DB_HOST=localhost
DB_PORT=3306
DB_NAME=cow_tracking
DB_USER=root
DB_PASSWORD=
```

> If your MySQL has a root password, put it in `DB_PASSWORD`.

Step 3 — Install dependencies

```
# Backend
cd backend
npm install

# Frontend
cd ../frontend
npm install
```

Step 4 — Create the database & sample data

```
cd backend
npm run mysql:migrate # creates the database and all tables
npm run mysql:seed    # adds test accounts and a sample cow
```

After seeding, these test accounts exist:

Role	Email	Password
Admin	admin@rwacow.rw	admin123
Vet	vet@rwacow.rw	vet123
Farmer	farmer@rwacow.rw	farmer123

Running the System

Open **two terminals**:

Terminal 1 — Backend

```
cd backend
npm run dev
```

You should see: `MySQL pool ready → localhost:3306/cow_tracking` and `Server running on port 5000`.

Terminal 2 — Frontend

```
cd frontend
npm run dev
```

You should see a local URL like `http://localhost:3000` (or `3001` if 3000 is busy).

Open that URL in your browser. 🌈

> **Keep both terminals and XAMPP MySQL running** while using the app. If the backend or MySQL stops, the app switches to a read-only "Demo Mode" with sample data.

User Guide

1. Welcome Page

The landing page explains the system and its features. Use the **EN / RW** toggle to switch language. Click **Get Started** to register or **Sign In** to log in.

2. Creating an Account

- Click **Create Account**
- Fill in full name, email, phone, and choose a role (Farmer, Vet, Government, or Slaughterhouse)
- Set a password (at least 8 characters)
- Click **Create My Account**

> Admin accounts cannot be self-registered — they are created by an existing administrator.

3. Signing In

- Enter your email and password
- Forgot your password? Click **Forgot password?**, enter your email, and follow the reset link

(in development, the reset link is printed in the backend terminal)

4. Dashboard

Shows live statistics:

- Total active cows, vaccinated, slaughtered, and sold
- Cows by district (bar chart)
- Upcoming vaccinations
- Scheduled slaughters

Click **Refresh** to reload the latest numbers.

5. Managing Cows

- **Cows** page lists all cows you're allowed to see
- Use **Search** and the **Status** filter to find cows
- Click **Register Cow** to add a new one

(farmers: the cow is automatically registered under your account)

- Click **View Details** on any cow to open its full profile

6. Cow Detail Page (Tabs)

- **Info** — breed, gender, color, weight, birth date, notes
- **Vaccinations** — history + next due date. *Vets/Admin/Government can add records; farmers view only.*
- **Offspring (Izakomotse)** — calves born from this cow
- **Transfers** — ownership history. *Admin/Government/Farmer can transfer.*
- **Slaughter** — slaughter record if any

7. Recording a Vaccination *(Vet / Government / Admin)*

- Open the cow → **Vaccinations** tab
- Fill in vaccine name, date given, next due date, and batch number
- Click **Save**

8. Registering Offspring

- Go to **Offspring** → **Register Offspring**
- Select the mother cow, enter the calf's tag, birth date, weight, and gender
- Click **Save** — the calf is automatically registered as a new cow

9. Transferring Ownership (Selling a Cow)

- Open the cow → **Transfers** tab
- Select the new owner, enter the sale price and date
- Click **Save** — the cow is now registered to the new owner

10. Slaughter Records *(Slaughterhouse / Government / Admin)*

- Go to **Slaughter** → **Schedule Slaughter**, pick a cow and date
- When done, click **Confirm Slaughter** and enter the meat weight

11. Language Switching

Use the **Kinyarwanda / English** button in the top bar at any time.

Data Privacy & Security

- **Passwords** are hashed with bcrypt — never stored in plain text
- **JWT tokens** protect every API request; expired/invalid tokens are rejected
- **Role-based access** is enforced on the server, not just hidden in the UI:

Data	Admin	Government	Vet	Farmer	Slaughterhouse
Cows	All	All	All	Own only	Scheduled only
Owners + National ID	Full	Full	No NID	Own only	✗

Data	Admin	Government	Vet	Farmer	Slaughterhouse
Vaccinations	✓	✓	✓	Own cows	✗
Offspring	✓	✓	✓	Own cows	✗
Slaughter	✓	✓	✗	✗	✓
Transfers + Prices	Full	Full	No prices	Own cows	✗
Database Backup	✓	✗	✗	✗	✗

A farmer can **never** see or modify another farmer's cows — this is enforced at the database-query level.



Backup & Export

Go to the **Export & Backup** page (visible to Admin, Government, Vet, Farmer).

- **Individual sheets** — download any table (cows, vaccinations, etc.) as an Excel `.xlsx` file
- **Full workbook** — all your permitted sheets in one Excel file
- **JSON backup** **(Admin only)** — a complete snapshot of the entire database

Each export is scoped to your role, so you only ever download data you're allowed to see.

Manual database backup (Admin/IT)

You can also back up MySQL directly:

```
C:\xampp\mysql\bin\mysqldump.exe -u root cow_tracking > backup.sql
```

Restore with:

```
C:\xampp\mysql\bin\mysql.exe -u root cow_tracking < backup.sql
```



Troubleshooting

Problem	Cause	Fix
Yellow "Demo Mode" banner	Backend is offline	Start backend (<code>npm run dev</code>) and make sure MySQL is running
"Registration failed"	Backend or MySQL not running	Start XAMPP MySQL, then the backend
MySQL won't start in XAMPP	Corrupted Aria log	Rename <code>aria_log.*</code> files in <code>C:\xampp\mysql\data</code> , then restart MySQL
"Invalid credentials" on login	Wrong email/password	Use a seeded account or reset the password

Problem	Cause	Fix	
Port 3000 in use	Another app is using it	Vite auto-switches to 3001 — check the terminal for the URL	
Port 5000 in use	Old backend still running	Find the PID with <code>`netstat -ano`</code>	<code>findstr :5000</code> and <code>taskkill /PID <pid> /F`</code>

Project Structure

```
Cow-management/
├── backend/                                # Node.js + Express API
│   ├── src/
│   │   ├── controllers/                  # Business logic (cows, owners, auth, etc.)
│   │   ├── routes/                      # API endpoint definitions
│   │   ├── middleware/                  # Auth & role checks
│   │   ├── database/                    # DB adapter, migrations, seed
│   │   ├── index.js                     # Server entry point
│   │   └── .env                         # Configuration
│   └── frontend/                        # React web app
│       ├── src/
│       │   ├── pages/                   # Each screen (Dashboard, Cows, Export, etc.)
│       │   ├── components/              # Shared UI (Layout)
│       │   ├── context/                 # Auth state
│       │   ├── i18n/                    # English + Kinyarwanda translations
│       │   └── services/                 # API client
│       └── README.md                    # This file
```

API Reference

Base URL: `http://localhost:5000/api`

All protected routes require an `Authorization: Bearer <token>` header.

Authentication

Method	Endpoint	Description
POST	<code>/auth/register</code>	Create an account
POST	<code>/auth/login</code>	Log in, returns a token
GET	<code>/auth/me</code>	Current user info
POST	<code>/auth/forgot-password</code>	Request a reset link
POST	<code>/auth/reset-password</code>	Set a new password

Cows

Method	Endpoint	Description
GET	<code>/cows</code>	List cows (filters: <code>status</code> , <code>search</code> , <code>district</code>)
GET	<code>/cows/stats</code>	Dashboard statistics
GET	<code>/cows/:id</code>	Full cow profile
POST	<code>/cows</code>	Register a cow
PUT	<code>/cows/:id</code>	Update a cow

Owners

Method	Endpoint	Description
GET	<code>/owners</code>	List owners
GET	<code>/owners/:id</code>	Owner detail + their cows
POST	<code>/owners</code>	Register an owner
PUT	<code>/owners/:id</code>	Update an owner

Vaccinations, Offspring, Slaughter, Transfers

Method	Endpoint	Description
GET/POST	<code>/vaccinations</code>	List / add vaccination
GET/POST	<code>/offspring</code>	List / register offspring
GET/POST	<code>/slaughter</code>	List / schedule slaughter
PUT	<code>/slaughter/:id/confirm</code>	Confirm slaughter
GET/POST	<code>/transfers</code>	List / transfer ownership

Export

Method	Endpoint	Description
GET	<code>/export/csv/:sheet</code>	Export a table as CSV
GET	<code>/export/backup</code>	Full JSON backup (admin only)



Support

For issues or questions about the system, contact your system administrator.

